

2024 SCHEME

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MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
FOURTH SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: PCCST402				
Course Name: DATABASE MANAGEMENT SYSTEMS				
Max. Marks: 60			Duration: 2 hours 30 minutes	
PART A				
		Answer all questions. Each question carries 3 marks	CO	Marks
1		Explain the Three-Schema Architecture of a DBMS with a neat diagram. How does it provide logical and physical data independence?	CO1	(3)
2		Distinguish between Super key, Candidate key, and Primary key. Illustrate with a real-world example from a university student registration system.	CO1	(3)
3		Define heuristic function. Write an SQL trigger on an Employees table that automatically logs the old and new salary values into an Audit_Log table whenever an employee's salary is updated. Give an example.	CO4	(3)
4		Explain the ER-to-Relational mapping rule for a Binary M:N Relationship Type. Illustrate with an example showing the resulting relational schema.	CO2	(3)
5		Explain the anomalies (insertion, deletion, update) that may occur in an unnormalized relation, with a suitable example.	CO2	(3)
6		Explain the Two-Phase Locking (2PL) protocol. Describe the growing and shrinking phases, and explain why 2PL guarantees serializability.	CO3	(3)
7		Explain the CAP theorem. For each pair of two properties (CA, CP, AP), name one real-world database system that prioritizes that combination and briefly justify.	CO5	(3)
8		Compare Key-Value Stores and Document Stores in terms of data model, query flexibility, and a practical use case for each.	CO5	(3)
PART B				
Answer any one full question from each module. Each question carries 9 marks				

Module 1				
9	a)	Explain centralized and client/server DBMS architectures with diagrams. Compare their advantages and limitations in the context of a modern enterprise application.	CO1	(5)
	b)	What are weak entity types? Explain with an example from the COMPANY database. Show how weak entities and their identifying relationships are represented in an ER diagram.	CO1	(4)
10		Draw an ER diagram for an Online Retail System based on the following requirements: - Manufacturers have a unique name, address, and phone number. - Products have a model number and a type. Each product is made by one manufacturer. Different manufacturers may have different products with the same model number, but no manufacturer has two products with the same model number. - Customers are identified by a unique social security number. They have email addresses and physical addresses. Several customers may live at the same address, but no two customers have the same email. - An order has a unique order number and a date. An order is placed by one customer. Each order contains one or more products, with a quantity for each product. Clearly identify all entities, relationships, attributes, keys, cardinality ratios, and participation constraints.	CO1	(9)
Module 2				
11	a	Consider the following schema: Suppliers (SID, sname, address) Parts (PID, pname, colour) Catalog (SID, PID, price) Catalog[SID] references Suppliers[SID]; Catalog[PID] references Parts[PID] Write relational algebra expressions for: - Find all prices for parts that are red or green. - Find the SIDs of all suppliers who supply a red part with price greater than 100. - Find the names of suppliers who supply every red part (use division).	CO2	(5)

	b	Explain the concept of Views in SQL. Write SQL statements to create a view that shows the supplier name and total number of parts supplied. Discuss one advantage and one security benefit of using views.	CO4	(4)
12		Consider the following database schema: Employees (EmployeeID INT PRIMARY KEY, FirstName VARCHAR(50), LastName VARCHAR(50), JobTitle VARCHAR(100), Salary DECIMAL(10,2), HireDate DATE, DepartmentID INT) Departments (DepartmentID INT PRIMARY KEY, DepartmentName VARCHAR(100), ManagerID INT) Foreign Key: Employees.DepartmentID references Departments.DepartmentID - Write SQL queries for the following: - Create both tables with appropriate constraints, including a CHECK constraint that Salary > 0. - Calculate the average salary per department and display only departments where the average salary exceeds 50,000. - List the employee name and department name for the employee with the highest salary in each department using a subquery. - Create an assertion that ensures no department has more than 100 employees. - Find employees hired in the last 2 years who earn more than the average salary of their department.	CO4	(9)
Module 3				
13	a)	Explain functional dependency with an example. How do functional dependencies help in identifying redundancy and guiding normalization?	CO2	(3)
	b)	A relation R(A, B, C, D, E) has the following functional dependencies: A → B, BC → D, D → E, A → C - Identify the candidate key(s). - Determine whether R is in 1NF, 2NF, and 3NF. Give reasons for each. - If R is not in 3NF, decompose it into 3NF relations.	CO2	(6)
14	a)	What is a conflict serializable schedule? Explain the precedence graph algorithm for testing conflict serializability. Determine whether the following schedule is conflict serializable. If yes, find an equivalent serial schedule:	CO3	(5)

		r1(X); r2(X); w1(X); r1(Y); w2(X); w1(Y)		
	b)	Compare the deferred update and immediate update database recovery protocols. Explain shadow paging as an alternative recovery technique and state when it is preferred.	CO3	(4)
Module 4				
15	a	Explain the BASE properties of NoSQL systems and compare them with ACID properties of relational databases. Use a real-world example (e.g., social media feed vs. banking transaction) to illustrate the trade-offs.	CO5	(4)
	b	Compare the four types of NoSQL architectural patterns: Key-Value Stores, Document Stores, Column Family Stores, and Graph Stores. For each, describe the data model and provide one industry use case.	CO5	(5)
16		A social media company needs to manage the following data: - User profiles (name, bio, settings, profile picture URL) - Posts (text, images, timestamps, likes count) - Friend/follower connections and recommendations - Real-time activity feed and notifications (a) For each data requirement, identify the most suitable type of NoSQL database and justify your choice. (5 marks) (b) Design the data structure/schema for any two of your chosen NoSQL databases, showing sample documents/records. (4 marks)	CO5	(9)

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